

Coastal Sands Textural Analysis from Ramakrishna Beach-Bhimunipatnam Tract (Visakhapatnam) East Coast of India

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Abstract:

This report includes textural analysis of Ramakrishna beach sands, including Mean size (Mz), Standard Deviation (SD), Skewness (SK), and Kurtosis (KG). Bhimunipatnam tract, Visakhapatnam East Coast, Andhra Pradesh. The textural properties of 23 typical samples from nine traverses of coastal sediments in distinct habitats (Backshore, Foreshore, Dune, Berm) between 17°46'-17°54' latitudes and 83°21'-83°27' longitudes were examined as part of this study. The samples for textural studies were separated into nine parts using sieves of various sizes (18,25,35,45,60,80,120,170,230). According to statistical research, the average values of coastal sediment are mean size (0.55 to 2.34), standard deviation (0.42 to 0.99), skewness (- 0.53 to 0.27), and kurtosis (- 0.53 to 0.27) (- 0.41 to 1.64). Scatter plots are used to determine the size parameters' interrelationships and geological importance. The nature of the sediments is predominantly unimodal, with medium sand being the main ingredient in diverse microenvironments such as backshore, foreshore, dune, and Berm, as shown by frequency distribution curves and scatter plots generated between different grain size parameters.

Keywords: Textural analysis, Backshore, Foreshore, Dune, Berm, East Coast.

Introduction:

PSDs are essential physical qualities of soil that are commonly expressed as a percentage of the total dry

weight of soil occupied by a certain size fraction. This feature is frequently used to classify soils and to estimate some hydraulic parameters (Campbell and Shiozawa, 1992). Grain size distribution is an important metric that sedimentologists have long used to deduce transport processes. The shape, size, and three-dimensional arrangement of the particles that make up sediment are referred to as sediment roughness. The composition of the source material, distance from the source and coast, geography and transit systems, and mechanical processes requiring extended wave action all influence grain size distribution. As a result, grain size analysis is critical in determining various sedimentological conditions (Mahadevan and Myrland, 1959, Folk and ward, 1957, Carver, 1971, Friedman, 1961 and 1967, Ramasastry and Myrland, 1959, Ramamohan Rao, et al., 1982, Bull, 1962, Bhat, et al., 2002, Jagannadha Rao and Krishna Rao, 1984, Dhanunjaya Rao, et al., 1989, Reddy and Malathi, 2002, Mohan and Rajamanickam, 1998, mohan, 2000, Nageswara Rao, et al., 2005, Rajesh et al., 2007, Ergine, et al., 2007, Ramanathan, et al., 2009, Rajasekhara Reddy and Karunakarudu, 2011, ganesh, et al., 2013, Karunakarudu et al., 2013). Furthermore, understanding the cause of the evolution of coastal sand ecosystems requires a study of particle size features. Rivers, rivulets, streams, waves and currents, sea level oscillations, shoreline configuration, winds, and other transportation and depositional agents, as well as aspects such as distance from the coast, distance from the source material, and character of the source material and topography of the area influence the grain size characters (Bangaku Naidu, et al., 2016). Earlier, many attempts have been made by several sedimentologists (Udden, 1914, Mason and Folk, 1958; Friedman, 1961, 1967). For Pudimadaka and Pentakota beaches, researchers looked at differences in grain size features of distinct beach habitats on the east coast and established connections between slope angle and mean grain size, concluding that the slope angle increases as grain size increases Sahu, 1964, Veerayya and Varadachari, 1975, Ramamohan Rao, et al., 1982, Jagannadha Rao and

Krishna Rao, 1984, Dhanunjaya Rao, et al., 1989, Frihy, et al., 1995, 2002, Hanamgond and Chavidi, 1997, Mohan and Rajamanickam, 1998, Reddy and Malathi, 2002, Nageswara Rao, et al., 2005, Rajasekhara reddy and Karunakarudu, 2009, Jagannadha Rao, et al., 2016, Mohan and Rajamanickam, (1998) employed grain size data to analyze the micro level changes in the depositional environments along the north of Pondicherry and Ennore beaches, concluding that some dune samples are unimodal and others are bimodal to polymodal sources of influence. Mohan (2000) investigated Quaternary landforms to assess variation in grain size distribution and depositional processes along the coast between Vedaranyam and Rameshwaram in Tamil Nadu. The grain size distribution of beach and sand dredged sands from the entrance of the harbor channel of Visakhapatnam, east coast of Andhra Pradesh, was compared by Mahadevan and Anjaneyulu (1954); Mahadevan and Nageswara Rao (1950). This may have been the first report of granulometry in beach sands from the Indian subcontinent. Later, Sastry et al., (1979) investigated configuration changes and fluctuation in grain size characteristics to better understand erosional and depositional processes along the Visakhapatnam-Bhimunipatnam coast, finding accretion from January to August and erosion from September to December. Chandramohan, et al., (1981) noticed erosion during the South West monsoon and deposition during the North East monsoon while studying the erosional and depositional ecosystems at Visakhapatnam beach (calm weather period). The current research focuses on the grain size distribution of coastal sands from Ramakrishna Beach to the Bhimili confluence, with the goal of better understanding sediment depositional conditions and patterns in the studied area.

Area of Research:

The area under inquiry is located between N 17° 46' and N 17°54' latitudes and longitudes E 83° 21' and E 83° 27' longitudes, and includes portions of Survey of India toposheet No: 65 O/6. The area south of Visakhapatnam,

Andhra Pradesh, India, stretches from Ramakrishna Beach to Bhimili (Figure 1). The Ramakrishna Beach-Bhimili tract resembles a basin, and it is flanked on the north, south, and west by the Kailasa, Yarada, and Narvahill ranges, respectively, and on the east by the Bay of Bengal. It features an undulating surface with elevations ranging from 4 to 50 meters above mean sea level (excluding hills).

The Study Area's Geology and Geomorphology:

The Precambrian Eastern Ghat Granulite Belt is characterized by garnet-sillimanite biotite gneiss (khondalites), hypersthene granites (charnockite), and garnetiferous granite (leptynite) (EGMB). Other rock types that appear as bedded and banded as well as enormous Formations include quartzite and pegmatite. Denudational hills ranging in height from 30 to 594 meters cover this area. Kailasa and Yarada extend roughly east to west, deviating from the Eastern Ghats' NE-SW pattern. Coastal and landward processes produced the geomorphic characteristics (Jagannadha Rao et al., 2012). The noteworthy features are the sediments, which reach a height of 30 meters above sea level. The Gosthani River is primarily responsible for the drainage pattern of the studied region.

Materials and Procedures:

By pushing a PVC tube (60 mm dia) down 9 Traverses from Ramakrishna Beach to Bhimili in varied microenvironments, such as dunes, back beach, fore beach, and berm, a total of 23 sediment samples are obtained. Coning and quartering are used to separate a representative part of the sediments weighing 100 grams from the bulk sediment samples. For 12 hours, 100 gram of sample is steeped in distilled water to dissolve the salts, followed by H_2O_2 to remove carbonates and HCl to remove shell material. The moist sediment is then dried in an oven at $1100^{\circ}C$. In a set of stacked sieves, dry samples are placed in the top sieve. Grain size analysis is performed on the samples using a conventional Ro-Tap sieve shaker at 12 ASTM mesh intervals (Hegde et al., 2006).

Seive Analysis:

In order to determine the grain size distribution of sand, sieving is widely utilized. For sieving analysis, we are using ASTM (American Society of Testing Materials) meshes. The dry sample is placed in the upper most sieve in a set of stacked sieves. After 10-15 minutes of shaking the sand that has formed on each sieve and the pan is removed and weighed, the stack of sieves is placed in Ro-Tap sieve shaker (the mesh numbers are 18, 25, 35, 45, 60, 80, 120, 170, 230, -230 accordingly). We calculated the weight percentage of each fraction in the sample and then the sample's overall weight percentage. On probability arithmetic graphs, the cumulative weight percentages were shown. As a result, ϕ 5, ϕ 16, ϕ 25, ϕ 50, ϕ 75, ϕ 84, and ϕ 95 values were achieved. According to the Folk and Ward formula, these data are utilized to compute textural parameters such as mean size (Mz), standard deviation (Sd), skewness (Sk1), and kurtosis (KG) (1957).

Results and Discussions:

According to the equations of Folk and Ward (1957), the data generated from the grain size analysis data was utilized to compute textural parameters such graphic mean size, graphic standard deviation, graphic skewness, and graphic kurtosis (Table1). In Backshore environments, the standard deviation varies from (0.44 ϕ to 0.74 ϕ) with mean size (1.02 ϕ to 2.3 ϕ). The skewness and kurtosis are varying from (-0.48 ϕ to 0.23 ϕ), (0.79 ϕ to 1.60 ϕ). In berm environment, the standard deviation varies from (0.4 ϕ to 0.98 ϕ) with mean size 1.3 ϕ the skewness and kurtosis are varying from (-0.18 ϕ to 0.15 ϕ), (0.48 ϕ to 0.68 ϕ). In the fore shore environment, the standard deviation varies from (0.5 ϕ to 0.99 ϕ) with mean size 1.25 ϕ the skewness and kurtosis are varying from (-0.53 ϕ to 0.57 ϕ), (0.41 ϕ to 1.36 ϕ). Table 1 shows the statistical analysis of beach sands of different microenvironments of study area, (1,2,3,4,5,6,7,8 and 9) are the station numbers. Meansize, standard deviation, skewness and kurosis of all microenvironment samples is in (ϕ) value.

Table 1. Statistical Analysis of Beach Sands of Different Microenvironments

Location	S. No	Environment	Meansize	Standard deviation	Skewness	Kurtosis
RKB	1	ForeShore	1.02	0.82	-0.14	0.41
SM	2	ForeShore	1.25	0.99	-0.12	0.62
VP	3	ForeShore	0.55	0.59	0.57	0.65
TP	4	ForeShore	1.24	0.59	-0.39	1.23
RS	5	ForeShore	1.77	0.71	0.02	1.01
INS	6	ForeShore	1.75	0.60	0.01	1.32
RB	7	ForeShore	2.34	0.59	-0.53	1.36
GR	8	ForeShore	1.27	0.80	0.01	0.89
BHI	9	ForeShore	2.2	0.56	0.12	0.87
R.K.B	1	BackShore	1.84	0.54	0.18	0.85
SM	2	BackShore	1.75	0.74	-0.19	0.93
VP	3	BackShore	1.02	0.56	-0.33	0.96
TP	4	BackShore	1.95	0.49	0.11	0.88
RS	5	BackShore	2.08	0.62	-0.14	0.79
INS	6	BackShore	2.3	0.50	-0.10	0.87
RB	7	BackShore	1.92	0.45	0.23	1.50
GR	8	BackShore	1.52	0.59	-0.48	1.64
BHI	9	BackShore	2.04	0.44	0.07	1.21
RK	1	Berm	1.64	0.42	0.15	0.48
SM	2	Berm	1.30	0.98	-0.18	0.55
VP	3	Berm	0.88	0.70	-0.18	0.68
INS	1	Dune	2.28	0.48	0.16	0.97
RB	2	Dune	2.05	0.55	0.27	0.82

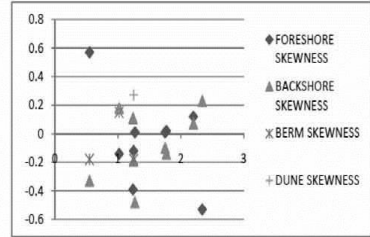
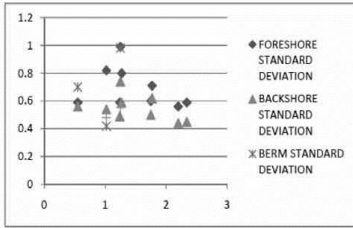


Figure 2(a). Mean size vs Standard Deviation

Figure 2(b). Mean size vs Skewness

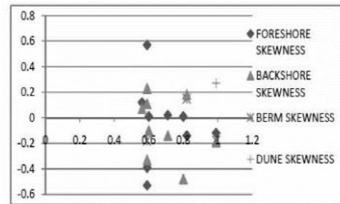
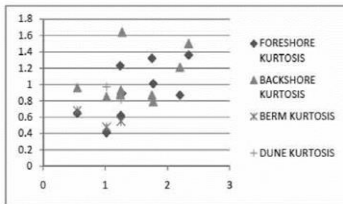


Figure 2(c). Mean size vs Kurtosis

Figure 2(d). Standard Deviation vs Skewness

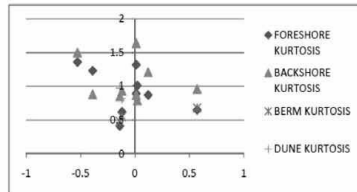
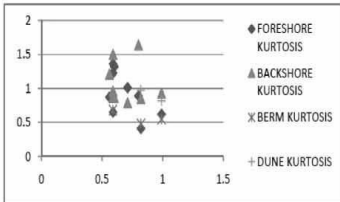


Figure 2(e). Mean size vs Kurtosis

Figure 2(f). Standard Deviation vs Skewness

Figure 2. (a, b, c, d, e, f) Scatter Plots

These trends and interrelationships in the scatter plots, according to Folk and Ward (1957) and others, could indicate the manner of deposition and hence aid in identifying the habitats. The scatter plot between mean size vs standard deviation (Figure 2a) of the present samples shows the clustering of values near the extreme end of right limb (Hough, 1942; Inman, 1949, 1953, Griffiths, 1951, Folk and Ward, 1957, Hubert, 1961, Seralathan and Padmalal, 1994).

The sediments are predominantly unimodal, with medium sand as the major element. The silt is secondary, resulting in a moderately sorted admixture. The scatter plot of mean size and skewness (Figure 2b) clearly shows the values, with some areas being positively skewed and others being negatively skewed (Folk and Ward, 1957). It also suggests that sediments are unimodal, with a higher proportion of medium sand and subordinate silt. The relationship between mean size vs kurtosis (Figure 2c) data shows that platykurtic has the upper hand, followed by mesokurtic. The plots between standard deviation vs skewness (Figure 2d) shows moderately sorted and positively negatively skewed sediments. The plot between stand deviation vs kurtosis (Figure 2e) shows that the majority of samples are of platy to mesokurtic nature, moderately sorted and of medium sand size. Plots of skewness vs kurtosis (Figure 2f) are effective tool in as certaining the textural aspects and interpreting the genesis of sediments as reflected in the normality of their size distribution.

Negatively skewed grain size curves, according to Duane (1964) and Cronan (1972), indicate erosion or non-deposition, whereas positively skewed curves suggest deposition. The findings show that the majority of the sediments have been well deposited. Non-deposition is indicated by some negatively skewed sediments. As a result, the positive skewness of these dements suggests that the grains are medium to fine. The findings show that the majority of the sediments have been well deposited. Non-deposition is indicated by some negatively skewed sediments. As a result, the positive skewness of these dements suggests that the grains are medium to fine.

Conclusion:

The beach sediments are sourced from the nearby hill ranges made of granulite facies of rocks that are part of the Precambrian Eastern Ghat Granulite Belt, according to textural studies data from Ramakrishna Beach - Bhimunipatnam tract, Visakhapatnam east coast (EGMB). The Kailasa Ranges and the Yarada Ranges are two

prominent hill ranges in this area. These hills range in elevation from 30 meters to 594 meters above sea level. The nature of the sediments is predominantly unimodal, with medium sand being the main ingredient in diverse microenvironments such as backshore, foreshore, dune, and Berm, as shown by frequency distribution curves and scatter plots generated between different grain size parameters. The sands of the dune environment are fine and very fine grained, dominantly well sorted followed by moderately sorted populations of positively skewed and platykurtic. Fine sands and negatively skewed sands, platykurtic, make up the sediments in the berm area. Fine and very fine sands cover the waterfront zones, which are equally populated by well sorted and moderately sorted nature, negatively skewed, leptokurtic to platykurtic sands.

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